DICE: Dual-Granularity Index for Claim-Enhanced Retrieval

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Abstract. Most retrieval methods are built on document-level representations; however, the nuanced differences between similar documents may be erased in densely compressed embeddings. This concern is particularly crucial in RAG, where the retriever frequently returns relevant documents with highly redundant information, undermining the overall effectiveness of generation. To reduce such redundancy, we propose a claim-based retrieval method, DICE, in which the claims are decomposed from documents by LLMs. DICE consists of two components: *claim-enhanced hybrid retrieval* and *claim-guided selection*, both of which leverage a claim-based index for efficiency. Our empirical results show average consistent 10% improvements in sub-topic (i.e., nugget)-level evaluation across two RAG benchmarks. Further analysis demonstrates that the claim-based index plays an important role in retrieving less-redundant documents. This exploration of claim-based retrieval shows its strength on practical agentic RAG applications. All the artifacts are open-sourced, and we anticipate claim-based retrieval to be applied more broadly to retrieval and beyond.[1]

Keywords: Claims, Granularity, Hybrid Retrieval, Diversity, K-Means, Clustering, RAG, MaxSim

1 Introduction

Retrieval-augmented generation (RAG) has emerged as a de facto search paradigm that aims to address complex information-seeking requests [21, 22], where users expect to see a system response that covers comprehensive relevant facts in a given corpus [16]. To evaluate such search systems, recent works have adapted to assess whether a system response contains required relevant facts (i.e., nuggets [31]). This shifts the goal of retrieval systems: Rather than providing a document list based on relevance, nugget coverage of the list becomes much more indicative [29]. Yet, typical retrieval design aims to score relevant documents as high as possible, which is subject to retrieving documents that are redundant in terms of nuggets, resulting in suboptimal effectiveness for RAG.

To alleviate this limitation, we integrate document retrieval with claim retrieval, where the retrieval unit is an atomic factual proposition extracted from

[1] https://anonymous.4open.science/r/claim-enhanced-retrieval/

documents. This design is particularly in line with how nuggets are defined in evaluation [31, 28, 32], and such fine-grained indexing has been shown to increase evidence recall in factoid QA tasks [7, 3]. However, whether claim-based retrieval can translate into coverage and how to utilize a claim index remain unclear.

In this work, we propose a training-free method: **D**ual-Granularity **I**ndexing for **C**laim-**E**nhanced Retrieval (DICE), which consists of two major components: (i) *Claim-enhanced hybrid retrieval* and (ii) *Claim-guided selection*. The retrieval is performed at both document- and claim-level indices and then combined into a list of document candidates. On top of the candidate list, we add a lightweight reordering process that greedily selects documents based on *estimated coverage gain* derived from the K-Means clusters over claims. Notably, both components reuse the claim-based index pre-computed offline with no query-time encoding or inference.

In our empirical evaluation, we conduct experiments on two TREC tracks that have the human-annotated nugget-level judgment: NeuCLIR [21] and RAGTIME [22]. We show that DICE achieves a 10% improvement on average compared to a Qwen3 embedding model across all coverage-based metrics; it achieves a 5% improvement over even a coverage-based retriever [17], demonstrating a promising improvement across different settings. Additionally, our ablation analysis provides insights into how the 'Claim' contributes to coverage effectiveness and insights into the interplay between relevance, diversity, and coverage.

2 Related Work

In RAG, retrieval controls access to external knowledge, so the generated response is bounded by the information the retrieved documents provide [16, 2]. Accordingly, recent benchmarks evaluate RAG systems with nuggets [28, 21, 29], which is an idea that traces back to definition QA [31] and summarization [27]. At the retrieval stage, these nuggets can be naturally integrated with IR metrics like subtopic recall [34] and α-nDCG [9], forming the coverage-based evaluation for retrieval systems [29], which is the target in this work.

Retrieval granularity has been discussed for decades [4] and is recognized as a key variable of retrieval effectiveness [37]. Recent research has explored indexing with a finer-grained level of retrieval units [23, 36]. This echoes NLP research in claim extraction [18, 26, 33], where text is decomposed into atomic and standalone statements [8]. For example, Chen et al. [7] observed that retrieving from decomposed propositions helps improve downstream QA accuracy. This work focuses on 'claim' only and leave the exploration of distinct decompositions [13, 6] as our future work.

To use finer-grained retrieval unit, the main challenge is to aggregate multiple claim retrievals while maximizing information coverage at the same time. For aggregation, a similar analogy appears in long-document retrieval [4] with pooling methods [11] or rank fusion [1, 3]. While maximizing coverage is similar to the goal of summarization, for example, Ernst et al. [14] generate summaries based on clustered propositions extracted from multiple documents. This aligns with

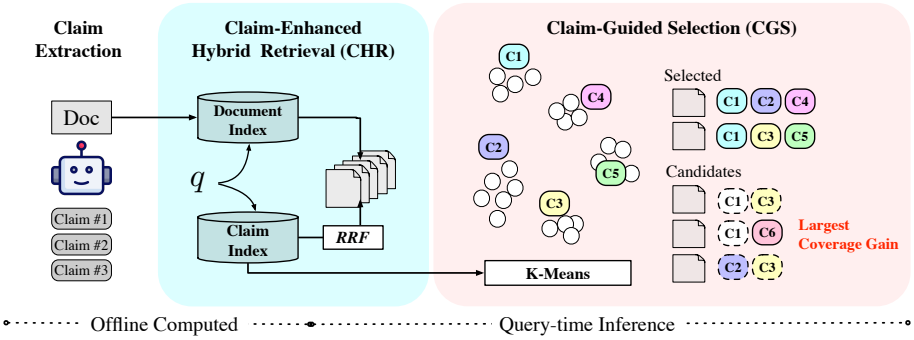


Fig. 1: Overview of DICE. Each document is first decomposed into atomic claims. Then, the documents and claims are pre-encoded as dual-granularity embedding indices. Followed by the two major components: **Claim-enhanced Hybrid Retrieval (CHR)**: A shared encoder retrieves from both indices, and the results are fused; **Claim-guided Selection (CGS)** with Clustering: A greedy method that selects the next documents based on the *estimated coverage gain*, considering overlap among selected clusters.

the idea of GINGER's generation pipeline [38], which conducts decomposition and clustering before response generation. DICE combines the above concepts all in one: it aggregates decomposed claim and document retrieved results, and explicitly optimizes nugget coverage based on claim pre-computed clusters.

3 DICE

3.1 Method Overview

Preliminaries. This study focuses on the effectiveness of coverage-based retrieval [16, 29], which evaluates retrieval systems based on the information coverage in a ranked list. Specifically, given a query, the retrieval system returns a ranked list of k documents $D = [d_1, \ldots, d_k]$ that aims to cover as many relevant nuggets as possible in the top-k documents. Such evaluation relies on subtopic-level judgments beyond the topic-level judgments used in ad-hoc ranking tasks. The ultimate goal is to seek comprehensive information for the query from a corpus $\mathcal{C}$.

Retrieval Architecture. We propose a training-free retrieval method, DICE, for improving coverage. As illustrated in Figure 1, it consists of multiple stages: Claim Extraction, Claim-Enhanced Hybrid Retrieval (CHR), and Claim-Guided Selection (CGS). Notably, the claim extraction can be conducted offline without adding query-time latency to retrieval. Each document in the corpus is decomposed into a list of atomic factual and self-contained statements [8]—claims. The decomposition is done by prompting an LLM (See Figure 2), the details of which are described in Section 4.

3.2 Claim-Enhanced Hybrid Retrieval (CHR)

First, we encode and index generated claims $c \in d \in \mathcal{C}$ into a claim-level index, and documents $d \in \mathcal{C}$ into a separate document-level index. As shown in Figure 1, both indices use the shared dense retrieval encoder, so a single query vector can be used to search with both indices.

Claim Retrieval with Pooling. When retrieving from the claim index, we apply a pooling step over the top-k_c retrieved claims. The pooling propagates each query-claim similarity $sim(c, q)$ to the document d they belong to, as the aggregated document relevance score. We denote the aggregated score as $s_{\text{claim}}(d, q)$, and then we rank documents by this score to obtain the final ranking list.

Sum. Sum pooling sums the similarity of the top-k_c retrieved claims, rewarding documents that have more similar claims. Document scores are computed as:

$$s_{\text{claim}}(d, q) = \sum_{c' \in d} sim(c', q), \quad \forall c' \in [c_1, ..., c_{k_c}]; \tag{1}$$

Reciprocal Rank Fusion (RRF). RRF pooling instead sums the reciprocal ranks of the retrieved claims. It only depends on the ranking of the retrieved claims when calculating the score, making it invariant to the scale of scores:

$$s_{\text{claim}}(d, q) = \sum_{c' \in d} 1/\big(\kappa + \text{rank}(\text{c}')\big), \quad \forall \text{c}' \in [\text{c}_1, ..., \text{c}_{\text{k}_\text{c}}], \tag{2}$$

where $\text{rank}(c)$ refers to the ranking of the claim. We set κ to 60 following standard practice [10]. The retrieval depth k_c is a hyperparameter that controls how many claims contribute to the final scores: a shallow depth may overly favor multi-claim documents, while a deeper one introduces noise from irrelevant documents. The detailed analysis is described in Section 5.2.

Hybrid Retrieval. We further combine the claim-derived document relevance scores with document-level relevance scores by linearly interpolating them:

$$s_{\text{CHR}}(d, q) = \lambda_1 \cdot s_{\text{claim}}(d, q) + (1 - \lambda_1) \cdot sim(d, q), \tag{3}$$

where $\lambda_1 \in [0, 1]$ balances the contributions of retrieved claims and the retrieved documents. For claim retrieval, we adopt the RRF as the default pooling strategy because it is more robust to noisy claims from irrelevant documents, as suggested in our analysis in Section 5.2.

3.3 Claim-Guided Selection (CGS)

Next, we introduce a document selection mechanism to score the top-ranked documents in a given ranking list D for enhancing information coverage. The fine-grained interactions between claims can be used to identify whether a document d_j contains more relevant information than the other candidates.

Estimated Coverage Gain. The selection greedily selects the document with the largest *estimated coverage gain*, that is, the document whose claims add the most information not yet covered by the documents selected so far. We refer to a selected list as D_s and denote the corresponding candidate list as $D_c = \{D \setminus D_s\}$. Formally, given a selected document list, we define the *estimated coverage gain* of a candidate document d_j as

$$\Delta(d_j|D_s) = \lambda_2 \cdot sim(q, d_j) + (1 - \lambda_2) \cdot \sigma\omega(d_j|D_s), \quad \forall d_j \in D_c, \tag{4}$$

where λ_2 controls the weights of document relevance and an adjustment term. The adjustment term has two parts: $\sigma \in \{+1, -1\}$ accounts for the direction; and the overlap function ω determines the information overlap between the targeted candidate document d_j and the selected document D_s. Notably, this formulation can generalize to distinct scenarios such as maximum marginal relevance (MMR) [5] when $\sigma = -1$ and the overlap function ω is document-level similarity.

Overlap function ω. The relationship between a candidate document and selected documents can be naturally computed by document-document similarity [5]. With claim index, we employ the MaxSim operator [19] to further capture the overlap via claim-level interactions:

$$\omega(d_j \mid D_s) = \max_{d_i \in D_s} \begin{cases} sim(d_i, d_j) & \text{(doc-doc similarity)}, \\ \dfrac{1}{|d_i|} \displaystyle\sum_{c \in d_i} \max_{c' \in d_j} \text{sim}(c, c') & \text{(claim-claim MaxSim)}. \end{cases} \tag{5}$$

Setting $\sigma = -1$ is analogous to MMR, whereas setting σ to 1 rewards the candidate document that obtains overlap with any of the selected documents. We found that it can instead expand the coverage effectively, aligning with the spirit of score regularization [12] and relevance feedback [20]. The detailed comparison is described in Section 5.3.

Refining overlap function ω^+. We further refine the overlap function with claim clustering over claims in candidate documents. Clustering with document-grounded claims yields a natural grouping of the unique information they contain, where each cluster serves as a proxy for a nugget: As shown in Figure 1, we first cluster all the claims in the top-ranked documents with K-means. Then, we represent each candidate document with a multi-hot vector with $\mathbf{k}$ dimensions as follows:

$$z(d_j) = \text{K-Means}(d_j, c \in [d_1, d_2, ..., d_m]) \in \{0, 1\}^{\mathbf{k}}, \quad \forall d_j \in D, \tag{6}$$

where m determines the depth of the considered documents.[2] And each element in $z(d)$ represents whether the document covers a specific claim cluster as depicted in Figure 1. For each candidate document d_j, we calculate the overlap by

[2] We set m to 20, as clustering documents that are deeper in the list may increase the risk of covering noise from irrelevant documents. We set $\mathbf{k}$ to 50; we found that values larger than 200 result in overly dispersed clusters and hurt overall performance.

discounting its already-covered clusters with an exponential decay according to how many selected documents have already covered it, such as

$$\omega(d_j \mid D_s) = \sum_{i=1}^{\mathbf{k}} \mathcal{I}(z(d_j), i)(1-\alpha)^n, \quad n = \sum_{d' \in D_s} \mathcal{I}(z(d'), i), \tag{7}$$

where $\mathcal{I}(z(d_j), i)$ identifies if document d_j contains the cluster i, so n stands for how many selected documents have already covered cluster i. The hyperparameter α determines the discount strength (0 means no discounts), mirroring the design in the α-nDCG [9]. It is worth noting that this claim-guided selection operates without additional inference since the required representations are pre-computed offline.

4 Experimental Setup

Nugget-based Evaluation. We evaluate on two TREC benchmarks with human-annotated nuggets and nugget-level relevance judgments:

- **NeuCLIR'24** [21] has 19 test topics and an average number of 20.9 nuggets per topic. The corpus consists of 10M English News articles. We encode each document together with its title and encode the query with the 'problem statement' field.
- **RAGTIME'25** [22] has 34 test topics and an average number of 15.6 nuggets per topic. The corpus consists of 4M English News articles.

Both English corpora are machine-translated from multilingual corpora and provided by the track organizers. We use StRecall and α-nDCG as our main metrics, which can showcase the coverage effectiveness by treating the nugget-level judgment as subtopics. The metrics are computed with `ir-measures` [24] toolkit.

Offline Claim Extraction. All the claims are extracted from the English versions of NeuCLIR'24 and RAGTIME'25 corpora. For each document, we prompt Llama-3.3-70B-Instruct [25] with the instructions (Figure 2) to generate a list of atomic claims, using a temperature of 0 for reproducibility. This results in 113.8M and 45.5M claims for NeuCLIR'24 and RAGTIME'25, respectively (about 11 claims per document on average).

Retrieval Methods. We describe baselines and methods below:

- **BM25 (doc)**[3] document retrieval using the original documents and its variants using the concatenated claims, **BM25 (claim)**;
- **CoveR** (ModernBERT-base, 140M) [17], which is pre-trained on coverage-based synthetic data, tailored for the coverage-aware retrieval task;

[3] https://github.com/xhluca/bm25s

Zero-shot Automatic Claim Extraction

System You are InformationExtractor, an intelligent assistant that extracts structured information from text by decomposing it into a list of standalone factual statements.

User
You will be given an article. Your task is to extract key statements from the article.

Instructions:
- Enclose each extracted statement with the tag '<s>' and '</s>'.
- Each extracted statements must appeared in the article.
- Ensure each statement is standalone and informative. Don't use any pronouns.
- Ensure the statements cover all the key points of the article.
- Only generate statements, do not include any additional texts or explanations.

News article:
- Title: $\{d_{\text{title}}\}$
- Content: $\{d_{\text{text}}\}$

<s>$\{c_1\}$</s><s> ... </s>

Fig. 2: The prompt for zero-shot claim extraction.

– **Qwen3-Emb** (0.6) [35] is a general-purpose embedding model transferred from a decoder architecture with large-scale instruction-tuning.

For fair comparisons, we use CoveR [17] and Qwen3-Embed [35] as the base embedder for DICE and its variants. Our empirical experiments explores

– **Claim-enhanced Hybrid Retrieval (CHR)**: retrieve from dual-granularity indices, and fuse claim retrieval with RRF pooling.
– **Claim-Guided Selection (CGS)**: reorder the given candidates based on document overlap. The candidates are dense retrieval results.
– **DICE (i.e., CHR + CGS)**: combining the hybrid retrieval and the claim-guided selection via claim clusters.

We also report two advanced LLM reranking methods:[4] **RankGPT** [30], a zero-shot LLM Listwise reranking method, and a coverage-based reranking method, **LANCER** [15], showing the improvement attributed to expensive cross-attention and query-awareness. We use Llama-70B [25] as the backbone for the two rerankers.

5 Empirical Results and Analysis

5.1 Main Results of DICE

Table 1 presents the evaluation results on NeuCLIR'24 and RAGTIME'25. DICE outperforms its base setting (**d** vs. **a**) across metrics, datasets, and the base embedding model used. With CoveR as the base embedding model, DICE improves StRecall by 2–8% and α-nDCG by 7–11%; the improvement is more prominent when using Qwen3-Embed as the base embedding model, up to 22.7% on NeuCLIR (e.g., +12.8 and +11.6 points in terms of α-nDCG@10 and @20). Moreover, we found that DICE can achieve comparable or even better performance

[4] The depth for CGS and reranking is 100.

Table 1: The coverage evaluation on 2 backbones with 4 compared systems: base retrieval, claim-enhanced hybrid retrieval (CHR), claim-guided selection (CGS) and both: DICE. Imprv. indicates the changes over the corresponding base condition (i.e., **a**). The best result is in bold. Superscripts denote that the system has significant gains over the noted system (paired t-test, $p < .05$).

		NeuCLIR				RAGTIME			
		StRecall		α-nDCG		StRecall		α-nDCG	
#	Condition	@10	@20	@10	@20	@10	@20	@10	@20
	BM25 (doc)	66.5	76.6	53.1	56.5	52.5	58.2	36.4	38.9
	BM25 (claim)	64.6	73.3	51.9	55.2	39.8	51.6	29.5	33.1
Base: CoveR – ModernBERT-base									
(a)	Base	68.8	78.9	57.1	60.0	65.9	73.6	52.3	54.9
(b)	w/ CGS	72.4	**82.5**	54.0	57.7	64.5	75.2	51.0	54.3
(c)	w/ CHR	69.9	79.4	**64.9**^{a}	**67.1**^{a}	69.3	**79.6**^{a}	**58.3**^{a}	**61.6**^{a}
(d)	w/ DICE	**73.8**	80.9	63.2^{b}	65.2^{b}	**71.4**^{br}	79.5^{a}	56.3^{b}	58.5^{b}
	Imprv. (%)	7.1	2.5	10.7	8.7	8.4	8.0	7.7	6.6
(l)	w/ LANCER	71.8	80.9	61.9	65.2	68.0	74.9	52.3	54.2
(r)	w/ RankGPT	66.7	76.8	53.0	56.7	65.4	76.3	56.2	59.6
Base: Qwen3-Embed-0.6B									
(a)	Base	69.6	78.0	56.4	59.6	69.6	77.0	56.0	58.4
(b)	w/ CGS	72.2	79.0	63.4^{a}	66.1^{a}	68.9	77.3	59.4^{a}	61.7^{a}
(c)	w/ CHR	74.0^{a}	82.6^{ab}	64.8^{a}	68.0^{a}	71.9	**80.2**^{ab}	57.5	60.5^{a}
(d)	w/ DICE	**78.4**^{abcr}	**84.5**^{abr}	**69.2**^{a}	**71.2**^{a}	**72.1**	79.6^{ab}	**61.2**^{al}	**63.5**^{al}
	Imprv. (%)	12.7	8.3	22.7	19.6	3.6	3.4	9.3	8.6
(l)	w/ LANCER	74.7	84.2	65.4	68.4	69.4	79.2	56.4	59.2
(r)	w/ RankGPT	70.8	80.8	63.8	66.6	70.6	79.5	57.2	60.1

than LLM reranking methods like RankGPT and LANCER, except for StRecall@20, highlighting the strong effectiveness of DICE.

To understand the effect of claim-guided selection (CGS), we compare DICE with CHR (**d** vs. **c**). Notably, with Qwen3-Embed, we see a clear improvement (α-nDCG@20 71.2 vs. 68.0 on NeuCLIR), with only a negligible drop in StRecall@20. However, CGS cannot improve a coverage-aware retriever. Adding CGS on top of either CoveR base retrieval or hybrid retrieval appears to have no significant difference, with only marginal improvement on StRecall and α-nDCG is degraded. We hypothesize this is due to the design of CoveR [17], which is fine-tuned to predict the coverage scores; thus, it breaks the estimation of coverage gain in Eq. (4). Comparing base retrieval with CGS (**a** vs. **b**) shows the similar observation; our empirical analysis in Section 5.3 also verifies this behavior.

Across two different base embedding models, we observe a similar improvement trend between claim-enhanced hybrid retrieval (CHR) and its base document retrieval (**c** vs. **a**). With CoveR, CHR significantly outperforms its baseline in terms of α-nDCG. With Qwen3-Embed, the improvement on NeuCLIR is also consistent but smaller on RAGTIME.

To summarize, claim-enhanced hybrid retrieval contributes the larger gain in DICE, while combining it with claim-guided selection can further boost the

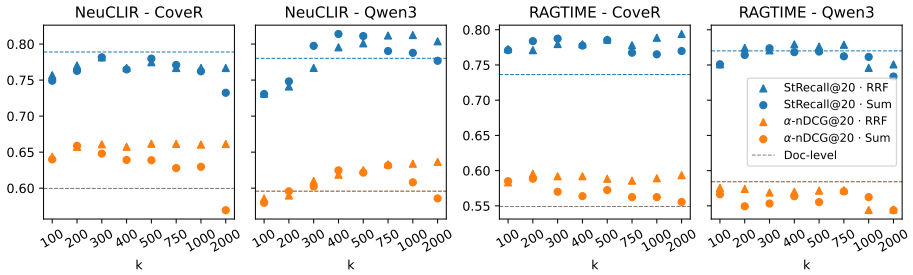


Fig. 3: The coverage evaluation results of claim retrieval with different pooling methods with respect to the depth of retrieved claims k_c. The dotted lines represent the effectiveness of document retrieval (baseline).

overall effectiveness with Qwen3-Embed, suggesting the two components provide complementary coverage-aware signals. However, with a base embedding model whose similarity scores are calibrated differently (e.g., CoveR), CGS improves only StRecall and slightly degrades the α-nDCG. Next, we analyze each component in detail in the following Sections.

5.2 Analysis of Claim-Enhanced Retrieval

Claim Retrieval. Figure 3 presents the results of claim-based retrieval with Sum and RRF Pooling. We found that there is no consistent gain across datasets and metrics; claim retrieval only slightly outperforms document retrieval (dotted line), and the effectiveness depends on the embedding model used. For example, CoveR's claim retrieval achieves better performance on RAGTIME across two metrics but degrades on NeuCLIR in terms of StRecall (blue marks in the first subfigure). On the other hand, Qwen3's claim retrieval achieves little improvement on NeuCLIR but degrades on RAGTIME in terms of α-nDCG.

Moreover, we found that the number of claims retrieved affects the effectiveness. Deeper depths ($k_c = 2000$) usually deteriorate coverage due to the risk of including noisy claims from irrelevant documents. This phenomenon is apparent in the Qwen3-Embed's results on both datasets. The degradation is more severe when using Sum pooling (circle marks), while RRF (triangle marks) alleviates this because it scores with the reciprocal ranks. Based on this finding, we use RRF pooling for DICE in the rest of the paper.

Hybrid Retrieval. To answer the question of whether document and claim retrieval are complementary, we compare retrieval using different indices: document-only, claim-only, and dual-granularity index. We then sweep the fusion weights λ_1 to examine the tension in the hybrid retrieval. As formulated in Eq. (3), the weights 0 and 1 mean relying solely on document or claim retrieval, respectively. We see that setting $\lambda_1 = 0.8$ consistently outperforms both claim-only and

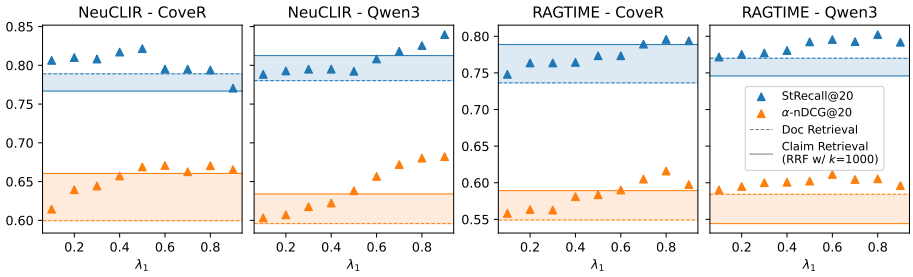


Fig. 4: Effectiveness of claim-enhanced hybrid retrieval with varying fusion weight λ_1. Blue and orange marks denote StRecall@20 and α-nDCG@20. Dashed and solid lines indicate document-only and claim-only retrieval, corresponding to effectiveness when $\lambda_1 = 0$ and 1; the shaded region spans the two cases.

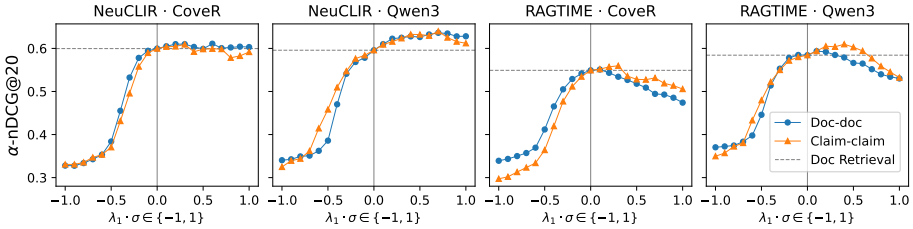


Fig. 5: A parameter analysis of adjustment direction and magnitude. We present them together on the same x-axis; the left and right halves of each subfigure indicate the different directions of the adjustment.

document-only retrieval (marks vs. the span), suggesting that the document and claim retrieval are complementary in terms of coverage effectiveness.

5.3 Analysis of Claim-Guided Selection

Impact of Adjustment on Coverage Effectiveness. According to the generalized formulation of *estimated coverage gain* in Eq. (4), Figure 5 systematically compare the adjustment term in multiple dimensions: (i) the granularity (i.e., doc-doc or claim-claim overlap, blue and orange dots, respectively) and (ii) the direction of adjustment (i.e., σ = {-1, 1}, left and right half on x-axis), and (iii) the magnitude of the adjustment (i.e., λ_2). In particular, when adjusting by penalizing overlap (e.g., MMR diversity reranking), we observe a downward pattern across all the evaluation results (i.e., the left half of subfigures). Penalizing candidate documents that are similar to the selected documents undermines the coverage regardless of the granularity, suggesting that the scale of adjustment scores outweighs the scale of relevance, causing irrelevant documents to be promoted to the top.

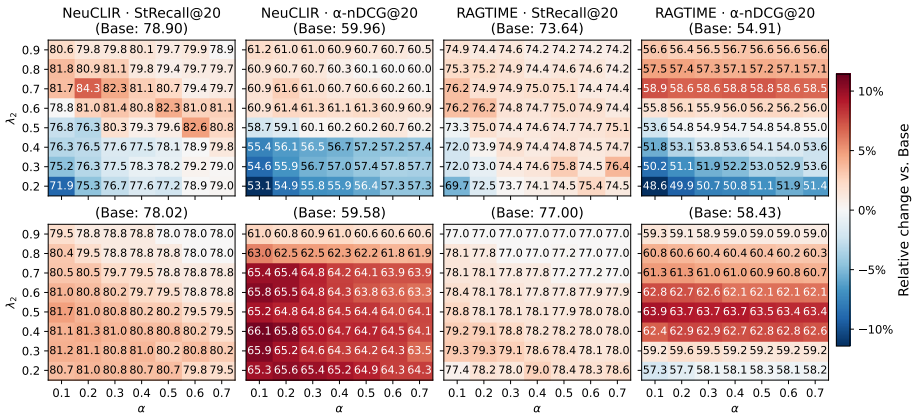


Fig. 6: Effect of α (x-axis) and λ_2 (y-axis) across four evaluation settings, using CoveR (top) and Qwen3-Embed (bottom) as the base embedding. Each cell shows the score; its color indicates the relative change from the original ranking (Base), with red for improvement, blue for degradation, and white for no change.

However, surprisingly, when adjusting by adding similarity ($\sigma = 1$, the right half), we found that the adjustment instead brings improvement in some cases, with more gain when using claim-level similarity scores. For example, on NeuCLIR and RAGTIME with Qwen3-Embed (the 2nd and 4th subfigures), the α-nDCG increases by 3 to 5 points when rewarding overlap. This shows that greedy selection favors documents that partially overlap with those already selected while still covering new subtopics, which is analogous to pseudo-relevance feedback [20] applied to subtopic coverage. Nevertheless, according to our empirical results, subtracting document-document similarity (MMR) is not an ideal strategy for improving coverage. However, rewarding partial overlap appears to be more crucial in coverage-based retrieval, which leads us to further refine the selection process with claims.

Claim-Guided Selection with Clusters. To understand the selection mechanism using claim clusters, we conduct a grid search over λ_2 and α. λ_2 controls the weight of relevance; as it increases, the selection process falls back to standard document retrieval. α determines the strength of the discount on covered clusters: a higher value discounts overlap clusters more aggressively.

As shown in Figure 6, claim-guided selection with clusters behaves differently depending on the base embedding model used: With CoveR, lowering λ_2 hurts performance dramatically (top heatmaps), whereas with Qwen3-Embed, lowering it improves performance (bottom). We hypothesize that CoveR's relevance scores already account for coverage because it's how it was trained [17], similar to findings in Section 5.1; thus, λ_2 should be set higher to avoid collapsing.

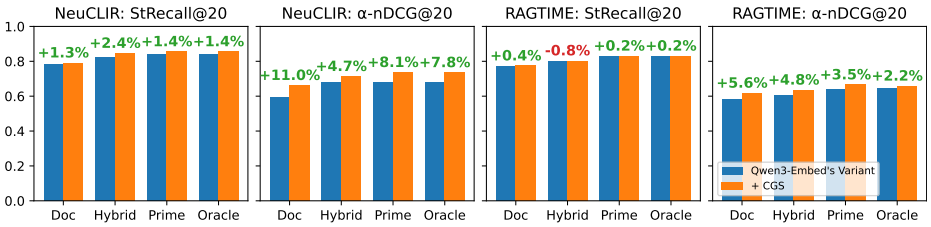


Fig. 7: Reordering improvement across different first-stage retrieved results and metrics on two benchmark datasets.

In contrast, the discount factor α behaves similarly across the two backbones. A light value is generally sufficient, as shown in the heatmap, where red indicates relative improvement over the base (original ranking); the best results typically are at $\alpha = 0.1$. Setting a discount factor too high over-decays the document that has covered clusters and therefore shifts the selection towards irrelevant clusters.

Robustness of First-stage Retrieval Figure 7 examines the robustness of claim-guided selection with clustering (CGS). We experiment with four retrieval environments, including (1) the general document retrieval (Doc); (2) the Claim-enhanced hybrid retrieval (Hybrid, Section 3.2); and (3) the *Prime* setting, which is the Doc's ranked list with only relevant documents, (4) the *Oracle* includes all the ground-truth relevant document. All of them use Qwen3-Embed embedding for computing similarity. Overall, CGS achieves consistent improvement across all settings, with an exception on RAGTIME's StRecall. Notably, the gain is biggest when adopting general document retrieval (the 1st bar), implying that standard document retrieval omits coverage awareness. When removing the effect of relevance (Prime and Oracle), we can still observe reasonable improvement (e.g., 8% and 3.5% of α-nDCG on NeuCLIR and RAGTIME), highlighting CGS's robustness across candidate lists.

6 Conclusion

In this work, we present DICE, a training-free method that aims to improve nugget coverage of retrieval. DICE leverages 'claim' and improves coverage in two ways: (1) hybrid retrieval that integrates two complementary signals–CHR, and (2) a document selection process with claim clusters–CGS. Empirical results demonstrate the promise of claim-enhanced retrieval, validating the potential of claims in improving information coverage. Our comprehensive analysis also reveals insights into balancing coverage and relevance, and provides empirical evidence of the pitfalls and directions of future retrieval research. The possible future work includes (i) integrating DICE to improve RAG report generation process, (ii) building claim-based corpus graphs for efficient knowledge retrieval, and (iii) investigating diverse claim extraction for more accurate selection.

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